THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 11: **Exploring Some Geometric Themes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A polyhedron has 12 vertices and 18 edges. The number of faces is: 1

(a) 8 **(b)** 6 **(c)** 28 **(d)** 32

2. A Koch curve starts with one segment. The number of segments at stage 1 is: 1

(a) 6 **(b)** 3 **(c)** 4 **(d)** 5

3. In the Sierpinski triangle (area of the original triangle taken as 1), the fraction of area still shaded after stage 2 is: 1

(a) $\frac{8}{16}$ **(b)** $\frac{9}{16}$ **(c)** $\frac{1}{16}$ **(d)** $\frac{6}{8}$

4. A cube of side 3 cm is painted on all faces and cut into 1 cm cubes. The number of small cubes with exactly one face painted is: 1

(a) 6 **(b)** 1 **(c)** 8 **(d)** 12

5. The net of a cylinder consists of: 1

(a) two circles and one rectangle **(b)** two circles and one triangle
(c) two rectangles and one circle **(d)** one circle and one sector

6. The top view of a cone standing on its base is a: 1

(a) square **(b)** circle (with a dot at the centre)
(c) rectangle **(d)** triangle

7. A polyhedron has 10 vertices and 15 edges. The number of faces is: 1

(a) 7 **(b)** 27 **(c)** 23 **(d)** 5

8. A Koch curve starts with one segment. The number of segments at stage 2 is: 1

(a) 9 **(b)** 8 **(c)** 15 **(d)** 16

9. In the Sierpinski triangle (area of the original triangle taken as 1), the fraction of area still shaded after stage 3 is: 1

(a) $\frac{27}{64}$ **(b)** $\frac{9}{12}$ **(c)** $\frac{1}{64}$ **(d)** $\frac{26}{64}$

10. A cube of side 4 cm is painted on all faces and cut into 1 cm cubes. The number of small cubes with exactly one face painted is: 1

(a) 24 **(b)** 96 **(c)** 8 **(d)** 25

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A prism has 21 edges. Find the number of sides of its base and the number of faces. 2

12. Verify Euler's formula for a nonagonal pyramid. 2

13. A solid has 9 faces and 21 edges. Use Euler's formula to find its number of vertices. 2

14. A prism has 27 edges. Find the number of sides of its base and the number of faces. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A cube of side 3 cm is painted on all faces and cut into 1 cm cubes. Find the number of small cubes with 3
(i) exactly two faces painted
(ii) exactly one face painted.

16. A Koch curve starts with a straight line of length 27 cm. How many segments are there at stage 3, how long is each, and what is the total length? 3

17. A Sierpinski triangle starts with a triangle of area 64 cm². Find the shaded area after stage 2 and the area removed. 3

OR

A cube of side 4 cm is painted on all faces and cut into 1 cm cubes. Find the number of small cubes with 3
(i) exactly two faces painted
(ii) exactly one face painted.

18. A Koch curve starts with a straight line of length 81 cm. How many segments are there at stage 4, how long is each, and what is the total length? 3

OR

A Sierpinski triangle starts with a triangle of area 256 cm². Find the shaded area after stage 2 and the area removed. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the number of faces, vertices and edges of 5
(i) a triangular prism and
(ii) a triangular pyramid, and verify Euler's formula for both.

OR

A cube of side 9 cm is painted and cut into 1 cm cubes. Find the number of cubes with three, two, one and no painted faces. Verify that these add up to 729. 5

20. Find the number of faces, vertices and edges of 5
(i) a square prism and
(ii) a square pyramid, and verify Euler's formula for both.

OR

A cube of side 10 cm is painted and cut into 1 cm cubes. Find the number of cubes with three, two, one and no painted faces. Verify that these add up to 1000. 5

END OF QUESTION PAPER — Total: 40 Marks